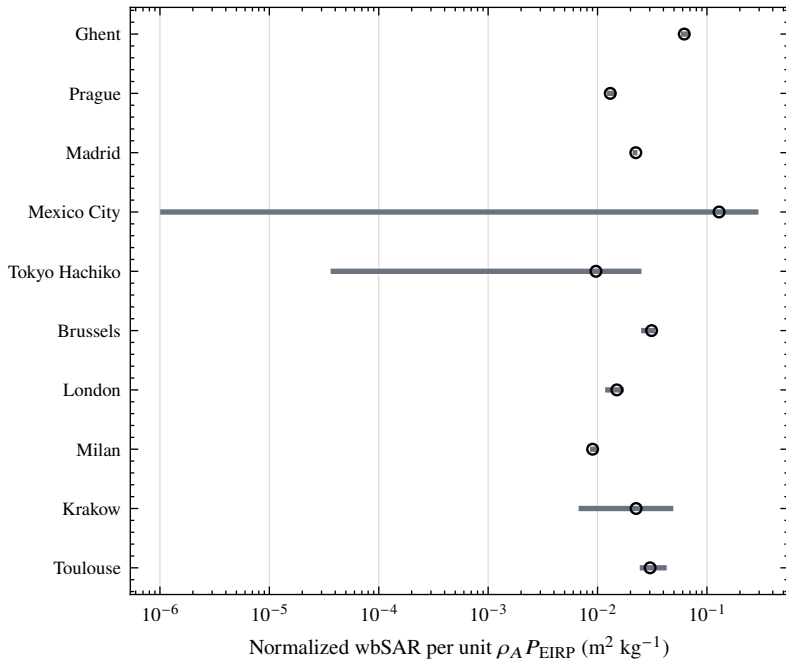


(a) Route q_{10} , q_{50} , and q_{90} 

(b) First-interaction components

